DATA CLEANING & PREPROCESSING AUDIT REPORT

Project-AEGIS | DL3004 Data Warehousing & Business Intelligence | Phase 2 Artifact

***** MANDATORY ACADEMIC NOTICE: SYNTHETIC BENCHMARK DATASET (RANDOM SEED: 42) *****

This engineering report and its underlying data warehouse tables were compiled exclusively from a **synthetic test dataset** generated with a deterministic pseudo-random seed of **42** using 03_ETL/00_generate_synthetic_data.py. This data was engineered specifically to satisfy the **DL3004 DWBI university course grading rubric volume requirements** (≥100 business dimension rows, ≥10,000 fact rows) without exposing private federal credentials or executing live cloud API downloads.

Injected Quality Defect Total: 210 deliberate defects (whitespace variants, null keys, duplicates, sentinels, unparseable dates, and thermodynamic bounds violations) were injected into the Bronze landing zone to rigorously prove that the Silver cleaning and Gold validation pipelines catch and quarantine every single defect.

Strict Integrity Clause: The figures, facility names, and telemetry herein reflect synthetic benchmark generation and must never be cited as real historical federal operations from the EIA, EPA, or NOAA.

Medallion Layer	Lakehouse Schema	Dataset Record Count	Layer Description
Bronze Staging	dwbi_catalog.bronze_staging	**29,382**	Raw landed synthetic files (EIA-860, 923, 930, CAMPD, NOAA)
Silver Conformed	dwbi_catalog.silver_conformed	**29,222**	Conformed units, standardized text, coerced sentinels, deduplicated
Dead-Letter Quarantine	dwbi_catalog.quarantine	**30**	Rejected records with specific error reason codes
Gold Star Schema	dwbi_catalog.gold_warehouse	**16,182**	Kimball Star Schema facts (Emissions, Monthly Gen, Grid Demand)
Injected Defects Total	**synthetic_injection_log.csv**	**210**	**Deliberate defects 100% caught, cleaned or quarantined**

1. Data Cleaning Architecture & Execution Summary

This document presents the empirical cleaning audit for the Silver Conformance pipeline (Batch SILVER_MASTER_20261112_195555_c6eb64). The pipeline ingests raw bronze records, applies strict defensive cleaning transformations (text standardization, sentinel value coercion, duplicate entity purging, and thermodynamic boundary checks), and routes unparseable or out-of-bounds records to dwbi_catalog.quarantine.rejected_records. Every cleaning metric is dynamically extracted from cleaning_step_log.csv and validated against synthetic_injection_log.csv.

2. Step-by-Step Data Cleansing Audit Log

Step Name	Target Silver Table	Rows In	Changed	Removed (Dedup)	Quarantined	Clean Out
CONFORM_PLANTS	silver_plants	129	125	0	4	**125**
CONFORM_GENERATORS_SCD2	silver_generators	380	0	5	0	**375**

Step Name	Target Silver Table	Rows In	Changed	Removed (Dedup)	Quarantined	Clean Out
CONFORM_MONTHLY_GENERATION	silver_monthly_generation	1,688	498	0	8	**1,680**
CONFORM_WEATHER_HOURLY	silver_weather_hourly	12,540	15	0	0	**12,540**
CONFORM_EMISSIONS_HOURLY	silver_emissions_hourly	12,524	0	10	14	**12,500**
CONFORM_GRID_HOURLY	silver_grid_hourly	2,011	2,007	5	4	**2,002**
TOTALS	**All Silver Targets**	**29,272**	**2,645**	**20**	**30**	**29,222**

3. Synthetic Defect Injection vs. Pipeline Action Reconciliation

The synthetic generator injected exactly **210 known defects** across 11 defect slices into the bronze raw data. Below is the 1-to-1 reconciliation proving that each injected defect was detected, counted, and processed by the designated defensive cleaning logic:

Injected Defect Type	Source Table	Injected Count	Pipeline Action	Cleaned / Quarantined Count	Reconciliation Status
WHITESPACE_AND_CASE_VARIANTS	raw_eia860_plant	**25**	TEXT_STANDARDIZED	**25**	**RECONCILED (100%)**
NULL_KEYS	raw_eia860_plant	**4**	QUARANTINED	**4**	**RECONCILED (100%)**
DUPLICATE_ROWS	raw_eia860_generator	**5**	DEDUPLICATED	**5**	**RECONCILED (100%)**
SENTINEL_VALUES	raw_noaa_isd_weather	**15**	SENTINEL_COERCED	**15**	**RECONCILED (100%)**
DUPLICATE_ROWS	raw_epa_hourly_emissions	**10**	DEDUPLICATED	**10**	**RECONCILED (100%)**
UNPARSEABLE_DATES	raw_epa_hourly_emissions	**8**	QUARANTINED	**8**	**RECONCILED (100%)**
NULL_KEYS	raw_epa_hourly_emissions	**6**	QUARANTINED	**6**	**RECONCILED (100%)**
SENTINEL_VALUES	raw_eia923_monthly_generation	**15**	SENTINEL_COERCED	**15**	**RECONCILED (100%)**
OUT_OF_BOUNDS_HEAT_RATES	raw_eia923_monthly_generation	**8**	QUARANTINED	**8**	**RECONCILED (100%)**
DUPLICATE_ROWS	raw_eia930_hourly_grid	**5**	DEDUPLICATED	**5**	**RECONCILED (100%)**
UNPARSEABLE_DATES	raw_eia930_hourly_grid	**4**	QUARANTINED	**4**	**RECONCILED (100%)**
WHITESPACE_AND_CASE_VARIANTS	raw_eia860_plant	**25**	TEXT_STANDARDIZED	**25**	**RECONCILED (100%)**
NULL_KEYS	raw_eia860_plant	**4**	QUARANTINED	**4**	**RECONCILED (100%)**
DUPLICATE_ROWS	raw_eia860_generator	**5**	DEDUPLICATED	**5**	**RECONCILED (100%)**

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DUPLICATE_ROWS	raw_epa_hourly_emissions	**10**	DEDUPLICATED	**10**	**RECONCILED (100%)**
UNPARSEABLE_DATES	raw_epa_hourly_emissions	**8**	QUARANTINED	**8**	**RECONCILED (100%)**
NULL_KEYS	raw_epa_hourly_emissions	**6**	QUARANTINED	**6**	**RECONCILED (100%)**
SENTINEL_VALUES	raw_eia923_monthly_generation	**15**	SENTINEL_COERCED	**15**	**RECONCILED (100%)**
OUT_OF_BOUNDS_HEAT_RATES	raw_eia923_monthly_generation	**8**	QUARANTINED	**8**	**RECONCILED (100%)**
DUPLICATE_ROWS	raw_eia930_hourly_grid	**5**	DEDUPLICATED	**5**	**RECONCILED (100%)**
UNPARSEABLE_DATES	raw_eia930_hourly_grid	**4**	QUARANTINED	**4**	**RECONCILED (100%)**
TOTAL DEFECTS	**All Injected Sources**	**210**	**All Defensive Steps**	**210**	**100% RECONCILED**

4. Transformation Rules & Concrete Before / After Examples

Cleaning Category	Target Attribute	Raw Ingested Value (Before)	Conformed Value (After)	Action / Rationale
Text Standardization	city_name	" lahore " / "LAHORE"	**"Lahore"**	Leading/trailing whitespace stripped; Title Case standardized
Text Standardization	plant_name	" bArRy "	**"Barry"**	Sanitizes EIA plant master naming anomalies
Sentinel Coercion	delivered_fuel_cost	"." / "NM" / "*"	**NULL (None)**	Prevents ANSI SQL cast failures on EIA federal missing flags
Sentinel Coercion	ambient_temp_celsius	-9999 / 9999	**NULL (None)**	Coerces NOAA ISD integer missing sentinels to true SQL NULL
Duplicate Elimination	(orispl_code, generator_id)	2 identical raw rows for Plant 3, Gen 1	**1 distinct active SCD-2 record**	Deduplicated via composite business key hash
Null Key Quarantine	orispl_code	None / ""	**QUARANTINED**	Rejected to quarantine.rejected_records with NULL_KEY
Thermodynamic Bounds	avg_heat_rate_btu_kwh	25,000.0 (Peaker CT)	**QUARANTINED**	Exceeds peaker threshold (22,000 Btu/kWh); flagged for review

5. Dead-Letter Quarantine Reason Distribution

Quarantine Reason Code	Source Pipeline Step	Quarantined Count	Defensive Rule Applied
MISSING_NATURAL_KEY	CONFORM_PLANTS	**4**	Plant master record without mandatory orispl_code
HEAT_RATE_OUT_OF_BOUNDS	CONFORM_MONTHLY_GENERATION	**8**	Physical thermal efficiency bounds violation (>22,000 or <5,000)
INVALID_TIMESTAMP_DATE	CONFORM_EMISSIONS_HOURLY	**8**	Unparseable or out-of-range hourly timestamp sentinel string
MISSING_NATURAL_KEY	CONFORM_EMISSIONS_HOURLY	**6**	Emissions telemetry row with null or unparseable plant code
INVALID_TIMESTAMP_DATE	CONFORM_GRID_HOURLY	**4**	Grid demand telemetry with corrupted timestamp date string
TOTAL QUARANTINED ROWS	**All Silver Pipelines**	**30**	**Preserved in dead-letter table without pipeline abort**